OLIST E-COMMERCE DATA ANALYTICS PROJECT

Consolidated KPI, Data Analysis & Business Insights Report

Project objective
Analyze the Olist Brazilian e-commerce dataset to understand customer behaviour, order and delivery performance, payment patterns, product/category sales, seller performance, and customer satisfaction. Translate the findings into practical business recommendations.

Report scope	Coverage
Customer & order behaviour	Customer records, unique customers, repeat purchases, order status
Delivery operations	Delivery time, early/late performance, state-wise delays
Payments	Payment methods, installments, payment value distribution and outliers
Commercial performance	Order items, products, categories, price and freight
Marketplace & experience	Seller performance/geography, reviews and response time
Data quality	Missing values, duplicates, unusual records and validation notes

Prepared from the analysis notes and results shared across the project conversations.

1. Executive Summary

The analysis indicates strong order fulfilment at an aggregate level, with 97.02% of orders marked delivered. However, repeat purchasing is low (3.12%), 8.11% of delivered orders with valid delivery dates were late, and customer and seller activity is geographically concentrated. Product and category sales show clear high-performing segments, while review and freight metrics point to opportunities in customer support and logistics.

Headline KPI	Result	Interpretation
Total order records	99,441	Overall order volume in the orders dataset
Delivered orders	96,478 (97.02%)	High delivered share
Repeat customer rate	3.12%	Customer retention is a major opportunity
Average actual delivery time	12.56 days	Median is 10.22 days; distribution has a long tail
Late delivery rate	8.11%	Based on 96,470 delivered orders with valid delivery dates
Average review score	4.09 / 5	Overall review sentiment is positive, but negative reviews need attention
Product revenue	R$13.59M	Sum of order-item product prices
Payment outlier rate	7.68%	IQR-flagged high-value payment records; not automatically errors

Interpretation caution: These metrics come from different tables and denominators. For example, payment records are not the same as unique orders, and customer records are not the same as unique customer identities.

2. Dataset & Data Quality Overview

Dataset	Rows × Columns	Key observations
Customers	99,441 × 5	No exact duplicate rows; 96,096 unique customer identities
Geolocation	1,000,163 × 5	261,831 exact duplicate rows; summarized to 19,015 ZIP-prefix records
Order items	112,650 × 7	98,666 unique orders; 32,951 unique products; 3,095 sellers
Order payments	103,886 × 5	One order can have multiple payment records
Order reviews	99,224 × 7	Large number of missing optional review-comment fields
Orders	99,441 × 8	Delivery timestamps have missing values; date fields converted for analysis
Products	32,951 × 9	610 missing values in several category/description fields; 2 missing in dimensions/weight fields
Sellers	3,095 × 4	Seller locations analysed by city/state
Category translation	71 × 2	Used to translate Portuguese category names into English

Geolocation exact duplicates were not blindly treated as invalid business records. ZIP-prefix aggregation was created using mean coordinates and record counts. Eleven ZIP prefixes (41 raw rows) were flagged outside approximate mainland Brazil coordinate bounds; Fernando de Noronha is a valid island exception. No source data was deleted during this exploratory stage.

3. Customer Analysis

KPI	Result
Customer records	99,441
Unique customer identities	96,096
One-time customers	93,099 (96.88%)
Repeat customers	2,997 (3.12%)
Orders from repeat customers	6,342

Geographic concentration: São Paulo (SP) contributes 41,746 customer records (41.98%), followed by Rio de Janeiro (12.92%) and Minas Gerais (11.70%). Top cities include São Paulo (15,540), Rio de Janeiro (6,882), and Belo Horizonte (2,773).

Business insight: Repeat purchase is relatively uncommon, while the customer base is concentrated in a small set of states and large cities. This suggests retention campaigns and regional growth initiatives are worth testing.

Recommended actions: loyalty/rewards, post-purchase engagement, personalized offers, repeat-purchase reminders, and targeted campaigns in underrepresented regions.

4. Orders & Delivery Performance

KPI	Result
Total orders	99,441
Delivered	96,478 (97.02%)
Shipped / canceled / unavailable	1,107 / 625 / 609
Valid delivered records used for timing	96,470
Average / median actual delivery	12.56 / 10.22 days
Early / late deliveries	88,644 / 7,826
Early / late share	91.89% / 8.11%
Average / median delay among late orders	9.55 / 5.81 days
Maximum observed late delay	188.98 days

Delay severity among late orders: 3,615 minor delays (46.19%), 2,827 moderate delays (36.12%), and 1,384 severe delays (17.68%). Approximately 53.8% of late orders were delayed by more than five days.

State-level findings: AL had the highest late-delivery percentage (23.93%; 95 of 397 orders). MA followed at 19.67%. Average delivery time was highest in RR (29.39 days), AP (27.19), and AM (26.43); SP was lowest at 8.76 days.

Business insight: Overall delivery is strong, but regional differences are material. The 188.98-day maximum is an extreme case that warrants investigation rather than being treated as typical performance.

Recommended actions: monitor logistics partners, investigate high-delay regions and extreme cases, improve estimated delivery predictions, and proactively notify customers about delays.

5. Payment Analysis

Payment analysis used both transaction-level and order-level views. Multiple payment records can belong to one order; therefore, payment transactions must not be interpreted as unique orders.

Payment method	Transactions	Payment value (BRL)	Average per transaction
Credit card	76,795	12,542,084.19	163.32
Boleto	19,784	2,869,361.27	145.03
Voucher	5,775	379,436.87	65.70
Debit card	1,529	217,989.79	142.57
Not defined	3	0.00	0.00

Payment method insight: Credit cards account for approximately 73.9% of payment records and are the dominant method. Boleto is the second most common.

Delivery status vs payment: Early-delivery orders had an average total order payment of R$158.72; late-delivery orders averaged R$172.73. This is an observed difference, not evidence that higher payments cause late delivery. The comparison used 88,644 early and 7,825 late orders with payment totals; one valid delivered order had no payment record.

Installments: Among credit-card records, one installment had the highest transaction count (25,455) and highest total payment value (R$2,440,445.43). Ten installments were also prominent, with 5,328 transactions and R$2,211,577.34 in payment value. Two zero-installment records were excluded from the positive-installment comparison, not deleted from the source data.

Payment value distribution: 103,886 records; mean R$154.10, median R$100.00, standard deviation R$217.49, maximum R$13,664.08, and skewness 9.25. This is a strongly right-skewed distribution.

IQR outlier metric	Result
Q1 / Q3	R$56.79 / R$171.84
IQR / upper fence	R$115.05 / R$344.41
Flagged payment records	7,981 (7.68%)
Credit card outlier rate	8.50%
Boleto / debit card / voucher rates	6.47% / 5.56% / 1.56%

The IQR lower fence was negative, so the flagged records were effectively high-value outliers. Outliers were retained. They may represent legitimate high-value purchases and should be investigated before any exclusion or fraud label is applied.

Recommended actions: monitor high-value transactions, assess fraud controls without over-flagging genuine customers, and continue tracking payment method mix and installment adoption.

6. Order Items & Product Performance

KPI	Result
Order-item rows	112,650
Unique orders in order items	98,666
Unique products	32,951
Unique sellers	3,095
Product revenue (price sum)	R$13,591,643.70
Freight value (sum)	R$2,251,909.54
Average product price	R$120.65
Average freight per order item	R$19.99
Average items per order	1.14
Maximum items in one order	21

Basket insight: 88,863 orders were single-item orders (90.06%); 9,803 were multiple-item orders (9.94%). Average items per order were 1.14. This suggests an opportunity to increase basket size.

Recommended actions: product bundles, related-product recommendations, buy-together offers, cross-selling, and carefully tested free-shipping thresholds.

7. Product Category Analysis

Category	Units sold	Revenue (BRL)	Revenue share
Health & Beauty	9,670	1,258,681.34	9.26%
Watches & Gifts	5,991	1,205,005.68	8.87%
Bed & Bath Table	11,115	1,036,988.68	7.63%
Sports & Leisure	8,641	988,048.97	7.27%
Computers & Accessories	7,827	911,954.32	6.71%
Furniture & Decor	8,334	729,762.49	5.37%
Cool Stuff	3,796	635,290.85	4.67%
Housewares	6,964	632,248.66	4.65%
Auto	4,235	592,720.11	4.36%
Garden Tools	4,347	485,256.46	3.57%

Health & Beauty is the highest-revenue category in the listed top 10. Bed & Bath Table leads by units sold. Watches & Gifts has the highest average product price among the top 10 categories (R$201.14). The top 10 categories account for approximately 62.36% of total revenue.

8. Best-Selling Products

Product ID (short)	Units sold
aca2eb7d	527
99a4788c	488
422879e1	484
389d119b	392
368c6c73	388
53759a2e	373
d1c42706	343

Product ID (short)	Units sold
53b36df6	323
154e7e31	281
3dd2a171	274

The top 10 products sold 3,873 units in total. High-demand products should be monitored for stock-outs; low-performing products may need targeted promotion. Product IDs are abbreviated here for readability.

9. Seller Performance & Geography

Seller KPI / finding	Result
Total sellers	3,095
Unique seller states	23
Unique seller cities	611? — unverified; confirm in notebook
Top revenue seller city	Guariba — R$229,472.63
Second by revenue	Lauro de Freitas — R$222,776.05
Top 5 seller states	SP 1,849 (59.74%); PR 349 (11.28%); MG 244 (7.88%); SC 190 (6.14%); RJ 171 (5.53%)
Top-10 seller detail	Ibitinga had the most products sold (1,987); Lauro de Freitas had highest average freight (R$31.90)

Approximately 59.74% of sellers are in São Paulo state; the top five states account for approximately 90.57% of sellers. Seller network concentration may create geographic coverage and resilience risks. High revenue does not necessarily mean highest sales volume.

Recommended actions: seller performance dashboards, stronger partnerships with high-performing sellers, investigation of high-freight sellers, and acquisition campaigns in underrepresented states. The seller city count is explicitly marked for validation because the source note did not confirm it.

10. Customer Reviews & Support Responsiveness

KPI	Result
Total reviews	99,224
Average / median review score	4.09 / 5
Positive / negative / neutral reviews	76,470 / 14,575 / 8,179
Positive / negative / neutral share	77.07% / 14.69% / 8.24%
Average / median response time	75.58 / 40.2 hours
Responses taking more than 7 days	4,914 (4.95%)
Maximum response time	518.7 hours

Most reviews are positive, but 14.69% are negative and warrant investigation. Average response time is approximately 3.15 days; a subset of responses took more than a week. Faster support and systematic complaint categorization may improve customer experience.

Data quality note: 814 repeated review_id records were identified, but they were not full duplicate rows. The same review_id appeared linked to multiple orders; records should not be deleted without understanding the relationship.

11. Price & Freight Analysis

Metric	Result
Price vs freight correlation	0.4142 (moderate positive relationship)
Price outliers	8,427 (7.48%)
Freight outliers	12,134 (10.77%)

Freight has a higher outlier share than product price. Higher-priced products may tend to have higher freight, but correlation does not establish causation. Product weight, dimensions, distance and seller location should be considered in deeper logistics analysis.

12. Consolidated Business Priorities

Priority area	Evidence from analysis	Suggested business response
Customer retention	Repeat customer rate is 3.12%	Loyalty program, personalized campaigns, post-purchase engagement
Basket size	90.06% single-item orders; 1.14 items/order	Bundles, cross-sell, related-product recommendations
Delivery reliability	8.11% late among valid delivered records; regional variation	Logistics partner scorecards, regional investigation, delay alerts
Customer experience	14.69% negative reviews; slow response tail	Complaint taxonomy, faster response SLA, root-cause analysis
Seller network	59.74% of sellers in SP; top five states 90.57%	Regional seller acquisition and network diversification
Product availability	Top 10 products sold 3,873 units	Demand-based inventory planning and stock-out prevention
Payment risk review	7.68% IQR outlier records	Review high-value payments with risk controls; retain legitimate transactions
Freight optimization	10.77% freight outliers; correlation with price 0.4142	Review shipping rates, dimensions, weight and route economics

13. Project Completion Status & Next Steps

Analysis area	Status from combined notes
Order items	Completed in the other analysis notes; KPI results included here
Products & categories	Completed
Customers	Completed
Orders & delivery	Completed
Reviews	Completed
Sellers	Basic analysis completed; unique seller city count needs verification
Geolocation	Exploratory analysis and ZIP-prefix summary completed
Category translation	Completed
Payments	Partially completed; major exploratory KPIs and outlier analysis included
Final dashboard	Pending
Final project report	This consolidated summary is a working report; final polish/validation remains

Recommended next steps: (1) validate seller-city count and any notebook outputs that were copied from the other analysis, (2) complete remaining payment questions if required, (3) build a consolidated dashboard with consistent definitions and filters, and (4) prepare a final project README and presentation.

14. KPI Definitions & Interpretation Notes

- **Customer repeat rate:** repeat customer identities divided by all unique customer identities (96,096).
- **Late delivery rate:** late deliveries divided by delivered orders with valid actual and estimated delivery timestamps (96,470).
- **Payment outlier rate:** IQR-flagged payment records divided by all payment records (103,886).
- **Product revenue:** sum of order_items price values; freight is reported separately.
- **Payment value:** sum of payment_value records; multiple payment records may belong to one order.
- **Review sentiment groups:** positive/neutral/negative classification as used in the supplied analysis notes; the precise score cut-offs should be documented in the notebook/report.

This report consolidates values supplied in the current project conversation and the uploaded analysis notes. It does not independently rerun the notebook. Figures marked for verification should be checked against the final executed notebook before publication.